

Beyond the Dyadic Fabius Web

Negative- q conjugacy, q -power decimation, sinc–Pochhammer factorization,
Thue–Morse desingularization, and reciprocal-radix limits

Research synthesis for the ProveIt Fabius-function corpus

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Abstract

The Fabius function and the Rvachev up-function sit at an unusual meeting point of smooth compactly supported probability densities, dyadic functional equations, Thue–Morse cancellation, infinite sinc products, Gaussian q -binomial coefficients, q -Pochhammer symbols, Bernoulli cumulants, Bell polynomials, and Lambert– W endpoint asymptotics. This report audits the current and archived LaTeX families in the **ProveIt** Fabius-function repository and then develops a single geometric-uniform probability family

$$X_q = (1 - q) \sum_{n \geq 0} q^n U_n, \quad U_n \stackrel{\text{iid}}{\sim} \text{Unif}[0, 1], \quad |q| < 1,$$

as a common organizing object.

Several exact results appear to be absent from the inspected corpus. For $0 < r < 1$, the negative-parameter law is not a genuinely new shape:

$$X_{-r} - \tfrac{1}{2} \stackrel{d}{=} \frac{1+r}{1-r} \left(X_r - \tfrac{1}{2} \right).$$

Consequently the variance-normalized and support-normalized laws at r and $-r$ coincide. Splitting digit indices modulo m gives a second exact identity,

$$X_q \stackrel{d}{=} \sum_{j=0}^{m-1} \frac{q^j}{[m]_q} X_{q^m}^{(j)},$$

so, in particular, the ordinary Fabius law is the convolutional barycenter of two independent quarter-base laws. Euler’s sine product then yields the sinc–Pochhammer factorization

$$\prod_{n \geq 0} \text{sinc} \left(\frac{(1-q)q^n t}{2} \right) = \prod_{k \geq 1} \left(\frac{(1-q)^2 t^2}{4\pi^2 k^2}; q^2 \right)_{\infty}.$$

This formula packages the zero set, spectral zeta function, Bernoulli cumulants, and the equality of normalized q and $-q$ profiles.

A further reciprocal-radix theorem incorporates the requested exterior values $q = \pm 2, \pm 4$. Normalized finite sums at b and b^{-1} have exactly the same law after reversing the iid digits. More strikingly, a desingularized finite Thue–Morse exponential polynomial is exactly the centered moment generating function of that finite convolution. Let

$$T_{m,b}(s) = \prod_{j=0}^{m-1} (1 - e^{b^j s}) = \sum_{n < 2^m} (-1)^{s_2(n)} e^{s A_b(n)}, \quad A_b(n) = \sum_{j=0}^{m-1} b^j b_j(n),$$

and $\alpha_{m,b} = (1-b)/(1-b^m)$. Then

$$\mathbb{E} e^{t(X_b^{[m]} - 1/2)} = \frac{(-1)^m e^{-t/2} T_{m,b}(\alpha_{m,b} t)}{\alpha_{m,b}^m b^{\binom{m}{2}} t^m}.$$

For $|b| > 1$, the left side converges to the centered transform of $X_{1/b}$. Thus $b = 2, -2, 4, -4$ produce, respectively, the laws at $q = 1/2, -1/2, 1/4, -1/4$. The report also proves log-concavity, an exact plateau phase for $|q| \leq 1/2$, reciprocal q -Lagrange row reversal, a position-weighted Thue–Morse/ q -Pochhammer master identity, and general-radix Bell–Bernoulli formulas. Conjectural sections formulate a general- q periodic Lambert– W transseries and a strict-log-concavity phase beyond the plateau threshold. A fully commented Python supplement verifies the exact and analytic identities and produces the figures.

Novelty and proof-status convention. CORPUS means that a result or its essential mechanism is already present in the inspected repository or cited literature. DERIVED HERE means that the statement was derived in this audit and was not located by exact-term, structural, and formula searches across the current canonical documents, consolidated frontier volumes, draft syntheses, and archive manifests. This is a claim of novelty relative to the inspected corpus, not a claim of global priority. CONJECTURAL marks statements for which only heuristics, numerical evidence, or a proof program is supplied. None of the newly derived claims is presented as Lean-formalized or peer-reviewed.

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1 Scope, corpus audit, and the non-duplication boundary

1.1 What was inspected

The recursive audit began with every current LaTeX-bearing family under <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs>. The principal sources were the formal exposition [1], the consolidated non-formalized frontier volume [2], the exponents-and- q -series synthesis [3], the Thue–Morse atlas and frontier synthesis [4], and the archive map [5]. The archive manifest was used to traverse historical families while treating byte-identical or explicitly consolidated copies as provenance rather than as independent mathematics. The repository’s paper collection and its surrounding citations were cross-checked against Arias de Reyna’s analytic and arithmetic papers [6, 7], Haugland’s dyadic evaluation paper [8], the digital and q -digital binomial theorems [9, 10], and later Thue–Morse work [11, 12].

The current primary exposition already contains a broad weighted-uniform-series API, the geometric law for $|q| < 1$, its affine recurrence, reflection, absolute continuity, and positive- q density bridges to Fabius and Rvachev. It also contains generic Gaussian-binomial and geometric Lagrange machinery, including explicit quarter-base cases. The current frontier register itself identifies negative-parameter density interpretation, the general- q Fourier route, centered transport, shape, moments, transforms, derivative hierarchies, and endpoint flatness as remaining boundaries [2]. The q -series synthesis develops general exponent-sequence sinc products, zero multiplicities, spectral zeta functions, cumulants, and q -Richardson ideas, while the Thue–Morse synthesis develops Boolean-cube products, Bell–Bernoulli moment formulas, Gaussian-binomial interpolation, and weighted digit enumerators [3, 4].

1.2 Results deliberately not counted as new

The following are used as foundations rather than advertised as discoveries:

- the identification of the half-base geometric-uniform law with the Fabius CDF and the Rvachev density;
- the ordinary Fourier product for the up-function and its finite-product approximants;
- standard q -Pochhammer and Gaussian-binomial identities, including the finite q -binomial theorem;
- generic cumulant/Bell-polynomial conversion and generic spectral-zeta formulas for exponent-sequence sinc products;
- the classical Thue–Morse product $\prod_{j < m} (1 - z^{2^j}) = \sum_{n < 2^m} (-1)^{s_2(n)} z^n$;
- the fixed-Hamming-weight Gaussian-binomial enumerator, which is closely related to the q -digital binomial theorem of Mansour and Nguyen [10].

1.3 Principal new-to-corpus seams

The report concentrates on interactions not found in the audit:

1. affine sign conjugacy for $q \mapsto -q$, including exact density, CDF, quantile, support, and inverse-function transport;
2. residue-class decimation $q \mapsto q^m$, especially the exact $1/2 \leftrightarrow 1/4$ convolution identity;
3. the double infinite sinc- q -Pochhammer product and its compatibility with decimation;
4. exact finite reciprocal-radix duality for $q \leftrightarrow q^{-1}$, giving a probability interpretation to $q = \pm 2, \pm 4$;

5. a desingularized Thue–Morse transform representation of finite and limiting geometric-uniform laws;
6. log-concavity and the exact plateau phase for the whole real family;
7. a two-variable digit-position master product whose negative- q specialization replaces Hamming parity by position parity;
8. general-radix Bell–Bernoulli formulas and centered $b \mapsto -b$ conjugacy;
9. a general- q Lambert- W endpoint program with exact negative- parameter transport.

2 The geometric-uniform family as a common language

2.1 Definition and elementary structure

Let $(U_n)_{n \geq 0}$ be independent $\text{Unif}[0, 1]$ random variables. For real $|q| < 1$, define

$$X_q = (1 - q) \sum_{n=0}^{\infty} q^n U_n. \quad (1)$$

The series converges absolutely almost surely. Its coefficients sum to one, so $\mathbb{E}X_q = 1/2$. Writing $V_n = U_n - 1/2$, one obtains the centered form

$$Y_q := X_q - \frac{1}{2} = (1 - q) \sum_{n=0}^{\infty} q^n V_n, \quad V_n \stackrel{\text{iid}}{\sim} \text{Unif}[-1/2, 1/2]. \quad (2)$$

The symmetry of each V_n gives $Y_q \stackrel{d}{=} -Y_q$, hence the law of X_q is symmetric about $1/2$ for every real $|q| < 1$, including negative q .

Separating the first digit gives the affine distributional fixed point

$$X_q \stackrel{d}{=} (1 - q)U + qX'_q, \quad (3)$$

where $U \sim \text{Unif}[0, 1]$, $X'_q \stackrel{d}{=} X_q$, and the two variables are independent.

Proposition 2.1 (Wasserstein uniqueness; DERIVED HERE). *For $|q| < 1$, the map*

$$\mathsf{T}_q(\nu) = \mathcal{L}((1 - q)U + qY), \quad Y \sim \nu,$$

is a contraction on probability measures with finite first moment in the 1-Wasserstein metric:

$$W_1(\mathsf{T}_q\mu, \mathsf{T}_q\nu) \leq |q|W_1(\mu, \nu).$$

Consequently $\mathcal{L}(X_q)$ is the unique finite-first-moment solution of (3).

Proof. Choose a coupling (Y, Z) of μ, ν , independent of the same uniform variable U . Then

$$\mathbb{E} |(1 - q)U + qY - ((1 - q)U + qZ)| = |q|\mathbb{E} |Y - Z|.$$

Taking the infimum over couplings gives the contraction estimate. Banach's fixed-point principle gives uniqueness. The partial sums in (1) converge to the fixed point, proving existence. $\square$

2.2 Support

For positive $q = r$, all coefficients in (1) are nonnegative and sum to one, so $\text{supp } X_r = [0, 1]$. For $q = -r$, positive coefficients occur at even indices and negative coefficients at odd indices. Their sums are

$$\sum_{j \geq 0} (1+r)r^{2j} = \frac{1}{1-r}, \quad -\sum_{j \geq 0} (1+r)r^{2j+1} = -\frac{r}{1-r}.$$

Thus

$$\text{supp } X_{-r} = \left[-\frac{r}{1-r}, \frac{1}{1-r} \right]. \quad (4)$$

The midpoint is still $1/2$.

2.3 Fabius and Rvachev normalization

Let F_q and f_q denote the CDF and density when these are defined. At $q = 1/2$, the repository's normalization gives

$$F_{1/2}(x) = F(x), \quad f_{1/2}(x) = 2 \text{up}(2x - 1), \quad (5)$$

where F is the bounded Fabius function and up is the standard Rvachev bump supported on $[-1, 1]$ [1, 6].

For general $q \neq 0$, it is convenient to use the support half-width

$$R_q = \frac{1-q}{2(1-|q|)} \quad (6)$$

and the support-normalized density

$$\mathcal{U}_q(x) = R_q f_q\left(\frac{1}{2} + R_q x\right), \quad -1 \leq x \leq 1. \quad (7)$$

At $q = 1/2$, $R_q = 1/2$, and (5) gives $\mathcal{U}_{1/2} = \text{up}$.

3 Negative q : exact affine sign conjugacy

3.1 The sign-conjugacy theorem

Theorem 3.1 (Affine sign conjugacy; DERIVED HERE). *Let $0 < r < 1$ and*

$$c_r = \frac{1+r}{1-r}.$$

Then

$$X_{-r} - \frac{1}{2} \stackrel{d}{=} c_r \left(X_r - \frac{1}{2} \right). \quad (8)$$

Equivalently, for the centered characteristic functions $\psi_q(t) = \mathbb{E}e^{it(X_q - 1/2)}$,

$$\psi_{-r}(t) = \psi_r(c_r t). \quad (9)$$

Proof. From (2),

$$Y_{-r} = (1+r) \sum_{n \geq 0} (-r)^n V_n.$$

Set $W_n = (-1)^n V_n$. Since the centered uniform law is symmetric, the variables W_n are again independent $\text{Unif}[-1/2, 1/2]$. Therefore

$$Y_{-r} \stackrel{d}{=} (1+r) \sum_{n \geq 0} r^n W_n = \frac{1+r}{1-r} \left((1-r) \sum_{n \geq 0} r^n W_n \right) \stackrel{d}{=} c_r Y_r.$$

Taking characteristic functions gives (9). □

This elementary sign change closes several negative-parameter gaps at once. It also explains why all even normalized invariants depend on q^2 .

Corollary 3.2 (CDF, density, and quantile transport; DERIVED HERE). *For $0 < r < 1$,*

$$F_{-r}(x) = F_r\left(\frac{1}{2} + \frac{x - 1/2}{c_r}\right), \quad (10)$$

$$f_{-r}(x) = \frac{1}{c_r} f_r\left(\frac{1}{2} + \frac{x - 1/2}{c_r}\right), \quad (11)$$

$$F_{-r}^{-1}(u) = \frac{1}{2} + c_r \left(F_r^{-1}(u) - \frac{1}{2}\right), \quad 0 < u < 1. \quad (12)$$

Every interior or endpoint asymptotic expansion for F_r , f_r , or the inverse CDF transfers exactly to $-r$ by these affine substitutions.

Proof. Equation (8) is an increasing affine transformation because $c_r > 0$. The CDF, density, and generalized inverse formulas follow by the ordinary change-of-variables rule. $\square$

Example 3.3 (The values $q = -1/2$ and $q = -1/4$). At $r = 1/2$, $c_r = 3$, so

$$F_{-1/2}(x) = F\left(\frac{x+1}{3}\right), \quad f_{-1/2}(x) = \frac{2}{3} \text{up}\left(\frac{2x-1}{3}\right), \quad (13)$$

with support $[-1, 2]$. At $r = 1/4$, $c_r = 5/3$, so

$$F_{-1/4}(x) = F_{1/4}\left(\frac{3x+1}{5}\right), \quad f_{-1/4}(x) = \frac{3}{5} f_{1/4}\left(\frac{3x+1}{5}\right), \quad (14)$$

with support $[-1/3, 4/3]$.

3.2 Normalized-shape collapse

The variance follows immediately from independent summation:

$$\sigma_q^2 := \text{Var}(X_q) = \frac{1}{12}(1-q)^2 \sum_{n \geq 0} q^{2n} = \frac{1-q}{12(1+q)}. \quad (15)$$

Corollary 3.4 (Variance-normalized collapse; DERIVED HERE). *Let $Z_q = (X_q - 1/2)/\sigma_q$. For $0 < r < 1$,*

$$Z_{-r} \stackrel{d}{=} Z_r. \quad (16)$$

Hence every standardized moment, cumulant, quantile, entropy shift, and shape statistic is an even function of q .

Proof. Equation (15) gives $\sigma_{-r} = c_r \sigma_r$. Divide (8) by this identity. $\square$

Corollary 3.5 (Support-normalized Rvachev profile; DERIVED HERE). *For $0 < r < 1$,*

$$\mathcal{U}_{-r} = \mathcal{U}_r. \quad (17)$$

Thus changing the sign of q changes only the physical location and scale; it does not create a new normalized bump.

Proof. For positive r , $R_r = 1/2$. For negative $-r$, $R_{-r} = c_r/2$. Substitute (11) into (7). $\square$

4 Residue-class decimation and the $1/2 \leftrightarrow 1/4$ bridge

4.1 The general decimation law

Write the q -integer

$$[m]_q = 1 + q + \cdots + q^{m-1} = \frac{1 - q^m}{1 - q}.$$

Theorem 4.1 (q -power decimation; DERIVED HERE). *Let $|q| < 1$ and $m \geq 1$. There exist independent copies $X_{q^m}^{(0)}, \dots, X_{q^m}^{(m-1)}$ such that*

$$X_q \stackrel{d}{=} \sum_{j=0}^{m-1} a_{m,j}(q) X_{q^m}^{(j)}, \quad a_{m,j}(q) = \frac{q^j}{[m]_q} = \frac{(1-q)q^j}{1-q^m}. \quad (18)$$

The coefficients satisfy $\sum_j a_{m,j}(q) = 1$, although some are negative when $q < 0$. Consequently

$$\phi_q(t) = \prod_{j=0}^{m-1} \phi_{q^m}(a_{m,j}(q)t), \quad \psi_q(t) = \prod_{j=0}^{m-1} \psi_{q^m}(a_{m,j}(q)t). \quad (19)$$

Proof. Split the series (1) by residue classes modulo m :

$$\begin{aligned} X_q &= (1-q) \sum_{j=0}^{m-1} \sum_{\ell \geq 0} q^{m\ell+j} U_{m\ell+j} \\ &= \sum_{j=0}^{m-1} \frac{(1-q)q^j}{1-q^m} \left((1-q^m) \sum_{\ell \geq 0} (q^m)^\ell U_{m\ell+j} \right). \end{aligned}$$

The parenthesized variables are independent copies of X_{q^m} . Their coefficients sum to one. Independence gives the characteristic products. $\square$

Corollary 4.2 (Cumulant compatibility). *For every $n \geq 1$ for which the cumulants are written,*

$$\kappa_n(q) = \kappa_n(q^m) \sum_{j=0}^{m-1} a_{m,j}(q)^n. \quad (20)$$

After inserting the Bernoulli formula from [section 7](#), this becomes the q -integer identity

$$\sum_{j=0}^{m-1} \left(\frac{q^j}{[m]_q} \right)^n = \frac{[m]_{q^n}}{[m]_q^n}. \quad (21)$$

4.2 Half base as a convolution of quarter-base laws

For $q = 1/2$, $m = 2$, the coefficients are $2/3$ and $1/3$:

$$X_{1/2} \stackrel{d}{=} \frac{2}{3} X_{1/4}^{(0)} + \frac{1}{3} X_{1/4}^{(1)}. \quad (22)$$

For $q = -1/2$, the same split gives

$$X_{-1/2} \stackrel{d}{=} 2X_{1/4}^{(0)} - X_{1/4}^{(1)}. \quad (23)$$

The first identity is a direct probabilistic bridge from the Fabius/Rvachev law to the quarter-base family already prominent in the repository. The second shows that the negative half-base law can be built from the same quarter-base blocks, with one reflected coefficient.

For a density g on a centered coordinate, define

$$(\mathcal{D}_a g)(x) = \frac{1}{|a|} g(x/a).$$

Because the support-normalized quarter-base profile is even, the sign in (23) disappears after normalization.

Corollary 4.3 (Rvachev convolutional barycenter; DERIVED HERE). *The standard Rvachev bump satisfies*

$$\text{up} = \mathcal{D}_{2/3} \mathcal{U}_{1/4} * \mathcal{D}_{1/3} \mathcal{U}_{1/4}. \quad (24)$$

The same right side is the support-normalized density of $X_{-1/2}$.

Proof. For positive q , the support radius is $1/2$, so support normalization turns (22) into $2(X_{1/2} - 1/2) = \frac{2}{3} 2(X_{1/4}^{(0)} - 1/2) + \frac{1}{3} 2(X_{1/4}^{(1)} - 1/2)$. Convolution gives (24). For $-1/2$, divide the centered version of (23) by its support radius $3/2$; the second coefficient becomes $-1/3$, which has the same effect as $+1/3$ on an even input density. $\square$

More generally, for every $m \geq 1$,

$$X_{1/2} \stackrel{d}{=} \frac{1}{2^m - 1} \sum_{j=0}^{m-1} 2^{m-1-j} X_{2^{-m}}^{(j)}. \quad (25)$$

At $m = 3$, the weights are $4/7, 2/7, 1/7$ and the building block is $X_{1/8}$.

5 Fourier products, smoothness, and endpoint flatness

5.1 The centered sinc product

For $V \sim \text{Unif}[-1/2, 1/2]$, $\mathbb{E} e^{itV} = \text{sinc}(t/2)$, where $\text{sinc } z = \sin z / z$. Hence

$$\psi_q(t) = \prod_{n=0}^{\infty} \text{sinc}\left(\frac{(1-q)q^n t}{2}\right), \quad |q| < 1. \quad (26)$$

The uncentered characteristic function is $\phi_q(t) = e^{it/2} \psi_q(t)$, and the fixed-point equation is

$$\psi_q(t) = \text{sinc}\left(\frac{(1-q)t}{2}\right) \psi_q(qt). \quad (27)$$

5.2 Superpolynomial Fourier decay

Theorem 5.1 (General real- q Fourier decay; DERIVED HERE). *Let $0 < |q| < 1$, put $r = |q|$ and $\lambda = -\log r$. Then, as $|t| \rightarrow \infty$,*

$$\log |\psi_q(t)| \leq -\frac{(\log |t|)^2}{2\lambda} + \mathcal{O}_q(\log |t|). \quad (28)$$

In particular, $|t|^N \psi_q(t) \rightarrow 0$ for every $N \geq 0$.

Proof. Set $A = (1-q)/2 > 0$. Since $|\text{sinc } x| \leq \min(1, |x|^{-1})$, choose $M = \lfloor \log(A|t|)/\lambda \rfloor$ when $A|t| > 1$. The first M factors in (26) give

$$\begin{aligned} \log |\psi_q(t)| &\leq -\sum_{n=0}^{M-1} \log(A|t|r^n) \\ &= -M \log(A|t|) + \frac{\lambda M(M-1)}{2}. \end{aligned}$$

Substituting $M = \log(A|t|)/\lambda + \mathcal{O}(1)$ yields (28). $\square$

Corollary 5.2 (Smooth density and flat endpoints; DERIVED HERE). *For every real q with $0 < |q| < 1$, the law of X_q has a compactly supported C^∞ density. Every derivative is obtained by Fourier inversion,*

$$f_q^{(k)}(x) = \frac{1}{2\pi} \int_{\mathbb{R}} (-it)^k e^{-itx} \phi_q(t) dt, \quad (29)$$

and all derivatives vanish at both endpoints of the support.

Proof. The decay in [theorem 5.1](#) makes every polynomial multiple of ϕ_q integrable, so repeated Fourier inversion gives a smooth density. A smooth function that is identically zero on an exterior open interval has all one-sided derivatives zero at the support endpoint; smoothness across that point makes every endpoint derivative vanish. $\square$

The case $q = 0$ is the uniform law and is intentionally excluded: its density is not smooth at the endpoints. This isolates a sharp regularity transition at $q = 0$.

6 The sinc- q -Pochhammer double product

6.1 Factorization theorem

Use the standard notation

$$(a; Q)_\infty = \prod_{n=0}^{\infty} (1 - aQ^n)$$

for the q -Pochhammer symbol [\[13\]](#). Euler's product $\text{sinc } z = \prod_{k \geq 1} (1 - z^2/(\pi^2 k^2))$ can be inserted into every factor of [\(26\)](#).

Theorem 6.1 (Sinc-Pochhammer factorization; DERIVED HERE). *For real $|q| < 1$,*

$$\boxed{\psi_q(t) = \prod_{k=1}^{\infty} \left(\frac{(1-q)^2 t^2}{4\pi^2 k^2}; q^2 \right)_\infty}. \quad (30)$$

The product converges locally uniformly in $t \in \mathbb{C}$, so it represents the same entire function as the sinc product.

Proof. Euler's product gives

$$\text{sinc}\left(\frac{(1-q)q^n t}{2}\right) = \prod_{k \geq 1} \left(1 - \frac{(1-q)^2 q^{2n} t^2}{4\pi^2 k^2}\right).$$

On every compact set in t , the double series of absolute first-order perturbations is bounded by a constant times $\sum_n |q|^{2n} \sum_k k^{-2} < \infty$. The products may therefore be reordered, and the product over n is exactly a q^2 -Pochhammer symbol. $\square$

Example 6.2 (The four interior values requested in the prompt). The factorization specializes to

$$\psi_{1/2}(t) = \prod_{k \geq 1} \left(\frac{t^2}{16\pi^2 k^2}; \frac{1}{4} \right)_\infty, \quad (31)$$

$$\psi_{-1/2}(t) = \prod_{k \geq 1} \left(\frac{9t^2}{16\pi^2 k^2}; \frac{1}{4} \right)_\infty = \psi_{1/2}(3t), \quad (32)$$

$$\psi_{1/4}(t) = \prod_{k \geq 1} \left(\frac{9t^2}{64\pi^2 k^2}; \frac{1}{16} \right)_\infty, \quad (33)$$

$$\psi_{-1/4}(t) = \prod_{k \geq 1} \left(\frac{25t^2}{64\pi^2 k^2}; \frac{1}{16} \right)_{\infty} = \psi_{1/4}\left(\frac{5}{3}t\right). \quad (34)$$

Thus $q = \pm 1/2$ share Pochhammer base $1/4$, while $q = \pm 1/4$ share base $1/16$. The sign changes only the spectral scale.

6.2 Decimation is Pochhammer dissection

The elementary dissection identity

$$(a; Q)_{\infty} = \prod_{j=0}^{m-1} (aQ^j; Q^m)_{\infty} \quad (35)$$

is the product-theoretic shadow of [theorem 4.1](#). Indeed, with $A_k = (1 - q)^2 t^2 / (4\pi^2 k^2)$,

$$\begin{aligned} \psi_{q^m}(a_{m,j}(q)t) &= \prod_{k \geq 1} \left(\frac{(1 - q^m)^2 a_{m,j}(q)^2 t^2}{4\pi^2 k^2}; q^{2m} \right)_{\infty} \\ &= \prod_{k \geq 1} (A_k q^{2j}; q^{2m})_{\infty}. \end{aligned}$$

Multiplying over j and using (35) gives exactly (30). Therefore residue-class splitting of the random digits and residue-class splitting of the Pochhammer product are not analogous operations; they are the same identity in probability and product coordinates.

7 Spectral zeta, Bernoulli cumulants, and Bell moments

7.1 The indexed zero spectrum

The positive zeros of the individual sinc factors occur at

$$\tau_{n,k} = \frac{2\pi k}{(1 - q)|q|^n}, \quad n \geq 0, \quad k \geq 1. \quad (36)$$

Coincident zeros are counted with their factor multiplicity. Define the indexed spectral zeta function

$$\mathcal{Z}_q(s) = \sum_{n \geq 0} \sum_{k \geq 1} \tau_{n,k}^{-s}.$$

Proposition 7.1 (Geometric spectral zeta; synthesis of corpus machinery). *For $\Re s > 1$,*

$$\mathcal{Z}_q(s) = \left(\frac{1 - q}{2\pi} \right)^s \frac{\zeta(s)}{1 - |q|^s}. \quad (37)$$

Near the origin,

$$\log \psi_q(t) = - \sum_{m \geq 1} \frac{\mathcal{Z}_q(2m)}{m} t^{2m}. \quad (38)$$

Proof. Sum the geometric progression over n and the Riemann zeta series over k . The logarithm of the canonical even product is $-\sum_{m \geq 1} t^{2m} \sum_{n,k} \tau_{n,k}^{-2m} / m$, proving (38) where the power series converges. $\square$

7.2 Bernoulli cumulants

The cumulant generating function of a uniform variable is

$$\log \frac{e^t - 1}{t} = \frac{t}{2} + \sum_{n=2}^{\infty} \frac{B_n}{n} \frac{t^n}{n!}, \quad (39)$$

with $B_1 = -1/2$. Independent weighted summation gives the following closed formula.

Theorem 7.2 (Bernoulli cumulants; corpus specialization completed for real q). *For $|q| < 1$,*

$$\kappa_1(X_q) = \frac{1}{2}, \quad \kappa_n(X_q) = \frac{B_n}{n} \frac{(1-q)^n}{1-q^n}, \quad n \geq 2. \quad (40)$$

All odd cumulants above the mean vanish. For $m \geq 1$, the even cumulants are also

$$\kappa_{2m}(X_q) = (-1)^{m+1} 2(2m-1)! \mathcal{Z}_q(2m). \quad (41)$$

Proof. The cumulant of order n of a weighted independent sum is the sum of the coefficient n -th powers times the input cumulant. Equation (39) gives $\kappa_n(U) = B_n/n$ for $n \geq 2$, and

$$\sum_{j \geq 0} ((1-q)q^j)^n = \frac{(1-q)^n}{1-q^n}.$$

Comparing the coefficient of t^{2m} in (38) with $\log \psi_q(t) = \sum_{m \geq 1} (-1)^m \kappa_{2m} t^{2m} / (2m)!$ gives (41). $\square$

Equation (40) may also be written

$$\kappa_n(X_q) = \frac{B_n}{n} \frac{(1-q)^{n-1}}{[n]_q}.$$

This is a direct Bernoulli–Gaussian- q -integer bridge.

7.3 Standardized cumulants and the Gaussian edge

From (15) and (40),

$$\kappa_{2m}(Z_q) = \frac{B_{2m}}{2m} 12^m \frac{(1-q^2)^m}{1-q^{2m}}. \quad (42)$$

This formula makes theorem 3.4 algebraically transparent. The first two non-Gaussian standardized cumulants are

$$\kappa_4(Z_q) = -\frac{6}{5} \frac{1-q^2}{1+q^2}, \quad (43)$$

$$\kappa_6(Z_q) = \frac{48}{7} \frac{(1-q^2)^2}{1+q^2+q^4}. \quad (44)$$

Proposition 7.3 (Gaussian limit; DERIVED HERE). *As $|q| \rightarrow 1^-$,*

$$Z_q \Longrightarrow N(0, 1). \quad (45)$$

As $q \rightarrow 0$, Z_q converges to the variance-one uniform distribution on $[-\sqrt{3}, \sqrt{3}]$.

Proof. By sign conjugacy it suffices to take $q = r \rightarrow 1^-$. The largest centered summand has magnitude at most $(1-r)/2$, while the standard deviation is asymptotic to $\sqrt{(1-r)/24}$. Their ratio tends to zero, so the bounded triangular array satisfies Lindeberg's condition. The second assertion follows directly from $X_q \rightarrow U_0$ in L^2 . $\square$

7.4 Bell-polynomial moments

Let $\mathcal{B}_n$ denote the complete exponential Bell polynomial, normalized by

$$\exp\left(\sum_{j \geq 1} x_j \frac{t^j}{j!}\right) = \sum_{n \geq 0} \mathcal{B}_n(x_1, \dots, x_n) \frac{t^n}{n!}.$$

Then

$$\mathbb{E}X_q^n = \mathcal{B}_n(\kappa_1(q), \dots, \kappa_n(q)), \quad (46)$$

and the centered moments are obtained by setting $\kappa_1 = 0$. Thus every moment is an explicit rational expression in Bernoulli numbers and the finite set of q -integers $[2]_q, \dots, [n]_q$. For rational q , all moments are rational.

8 Shape theory: log-concavity and the plateau transition

8.1 Log-concavity for every real parameter

Theorem 8.1 (Log-concavity; DERIVED HERE). *For every real q with $0 < |q| < 1$, the density f_q is log-concave on its support. It is symmetric about $1/2$, positive on the support interior, and unimodal.*

Proof. Every finite partial sum in (2) is a convolution of uniform densities on intervals, possibly reflected. Uniform interval densities are log-concave, and convolution preserves log-concavity by the Prékopa–Leindler theorem. The partial-sum laws converge weakly to $\mathcal{L}(Y_q)$; the class of log-concave probability measures is closed under weak convergence. By theorem 5.1, the limit has a smooth density. A nonzero log-concave density is positive on the interior of its interval support. Symmetry was proved from the centered series, and a symmetric log-concave density is unimodal at the center. $\square$

8.2 Exact plateaus for $|q| \leq 1/2$

For $q = r > 0$, the fixed point (3) is a convolution of rX'_r with a uniform interval of length $1 - r$. Therefore

$$f_r(x) = \frac{1}{1-r} \left[F_r\left(\frac{x}{r}\right) - F_r\left(\frac{x - (1-r)}{r}\right) \right], \quad (47)$$

where the CDF is extended by 0 and 1 outside $[0, 1]$.

Theorem 8.2 (Plateau phase; DERIVED HERE). *Let $0 < r \leq 1/2$.*

1. *For positive parameter,*

$$f_r(x) = \frac{1}{1-r} \quad \text{for } x \in [r, 1-r], \quad \|f_r\|_\infty = \frac{1}{1-r}. \quad (48)$$

2. *For negative parameter,*

$$f_{-r}(x) = \frac{1}{1+r} \quad \text{for } x \in \left[\frac{r^2}{1-r}, 1 - \frac{r^2}{1-r} \right], \quad \|f_{-r}\|_\infty = \frac{1}{1+r}. \quad (49)$$

When $r < 1/2$, the displayed intervals have positive length. At $r = 1/2$, each plateau collapses to the single central maximizer.

Proof. If $x \in [r, 1 - r]$, then $x/r \geq 1$ and $(x - (1 - r))/r \leq 0$. Equation (47) therefore equals $1/(1 - r)$. Convolution with a probability density and a uniform kernel gives the matching global upper bound. The negative statement is the affine image under (11); the interval endpoints simplify to $r^2/(1 - r)$ and its reflection, and the height is divided by c_r . $\square$

The exact plateau at $q = 1/4$, of height $4/3$ on $[1/4, 3/4]$, and its negative counterpart, of height $4/5$ on $[1/12, 11/12]$, are visible in figure 1.

Conjecture 8.3 (Strict log-concavity beyond the threshold; CONJECTURAL). *For $1/2 < r < 1$, the support-normalized density $\mathcal{U}_r$ is strictly log-concave on $(-1, 1)$. Equivalently, the only obstruction to strictness in the real geometric-uniform family is the exact plateau phase $0 < |q| < 1/2$ and its critical collapse at $|q| = 1/2$.*

The conjecture is consistent with the numerical profiles and with the Gaussian limit, but a proof appears to require a strict form of convolution log-concavity adapted to infinitely many nested uniform scales.

9 Reciprocal q : bringing $\pm 2, \pm 4$ into the picture

9.1 Finite Pochhammer and Gaussian reciprocity

For $q \neq 0$, finite products satisfy

$$(a; q)_n = (-a)^n q^{\binom{n}{2}} (a^{-1}; q^{-1})_n. \quad (50)$$

The Gaussian polynomial consequently obeys

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = q^{k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_{q^{-1}}. \quad (51)$$

For the nodal polynomial $P_n^{(q)}(z) = \prod_{j=1}^n (z - q^j)$,

$$P_n^{(q)}(z) = (-1)^n q^{n(n+1)/2} z^n P_n^{(q^{-1})}(z^{-1}). \quad (52)$$

Let $\lambda_{n,k}^{(q)}$ be the Lagrange weight for evaluating a polynomial at zero from its values at $1, q, \dots, q^n$. The standard closed form is

$$\lambda_{n,k}^{(q)} = \frac{(-1)^{n-k} q^{\binom{n-k+1}{2}}}{(q; q)_n} \begin{bmatrix} n \\ k \end{bmatrix}_q. \quad (53)$$

Theorem 9.1 (Reciprocal row reversal; DERIVED HERE). *For every admissible $q \neq 0$,*

$$\lambda_{n,k}^{(q)} = \lambda_{n,n-k}^{(q^{-1})}. \quad (54)$$

Hence the rows for $q = 2, -2, 4, -4$ are exact reversals of the rows for $q = 1/2, -1/2, 1/4, -1/4$, respectively.

Proof. The geometric node list for q^{-1} , read backwards, is $q^{-n}, q^{-(n-1)}, \dots, 1$, which becomes the q -node list after a common nonzero scaling by q^n . Lagrange weights for evaluation at zero are unchanged by common scaling and are reversed with the node order. Alternatively, substitution of (51) into (53) gives the same identity algebraically. $\square$

9.2 Finite normalized geometric laws

The infinite probability series (1) does not converge for $|q| > 1$. There is nevertheless a canonical finite normalization:

$$X_q^{[m]} = \alpha_{m,q} \sum_{j=0}^{m-1} q^j U_j, \quad \alpha_{m,q} = \frac{1-q}{1-q^m}, \quad (55)$$

provided $q \neq 1$ and $q^m \neq 1$. The coefficients sum to one, although negative q produces signed affine coefficients.

Theorem 9.2 (Exact reciprocal-radix law; DERIVED HERE). *For every real admissible $q \neq 0$,*

$$X_q^{[m]} \stackrel{d}{=} X_{q^{-1}}^{[m]}. \quad (56)$$

If $|q| > 1$, then

$$X_q^{[m]} \implies X_{q^{-1}} \quad (m \rightarrow \infty). \quad (57)$$

Thus the exterior parameters $2, -2, 4, -4$, interpreted through normalized finite sums, converge to the interior laws at $1/2, -1/2, 1/4, -1/4$.

Proof. A direct calculation shows

$$\alpha_{m,q} q^{m-1-j} = \alpha_{m,q^{-1}} q^{-j}.$$

Reversing the iid list $(U_0, \dots, U_{m-1})$ therefore turns $X_q^{[m]}$ into $X_{q^{-1}}^{[m]}$ without changing its law. If $|q| > 1$, then $|q^{-1}| < 1$, and the finite truncations on the reciprocal side converge almost surely and in every L^p to $X_{q^{-1}}$. $\square$

This theorem is the probability counterpart of Pochhammer reciprocity and Lagrange row reversal. It also prevents a common conceptual mistake: values $|q| > 1$ should not be inserted naively into the infinite series, but they do possess exact finite and limiting meanings after normalization and digit reversal.

10 A Thue–Morse desingularization theorem

10.1 Signed subset exponential transforms

Let $b_j(n) \in \{0, 1\}$ be the binary digits of $n < 2^m$, let $s_2(n) = \sum_j b_j(n)$, and put $\varepsilon_n = (-1)^{s_2(n)}$. For a real radix b , define the signed digit evaluation

$$A_b(n) = \sum_{j=0}^{m-1} b^j b_j(n). \quad (58)$$

The finite exponential transform is

$$T_{m,b}(s) = \prod_{j=0}^{m-1} (1 - e^{b^j s}) = \sum_{n=0}^{2^m-1} \varepsilon_n e^{s A_b(n)}. \quad (59)$$

For $b = 2$, $A_2(n) = n$, so this is the classical Thue–Morse exponential polynomial. For $b = -2$, it evaluates the same binary digit subsets in negabinary place values; $b = \pm 4$ produces sparse base-four and negative base-four evaluations.

The product in (59) has a zero of exact order m at $s = 0$, the analytic form of Prouhet cancellation. Dividing out that zero produces not merely a convenient regularization, but a probability transform.

Theorem 10.1 (Thue–Morse desingularization; DERIVED HERE). *Let $b \in \mathbb{R} \setminus \{0, 1\}$, assume $b^m \neq 1$, and set $\alpha_{m,b} = (1 - b)/(1 - b^m)$. Then*

$$\boxed{\mathbb{E}e^{t(X_b^{[m]} - 1/2)} = \frac{(-1)^m e^{-t/2} T_{m,b}(\alpha_{m,b} t)}{\alpha_{m,b}^m b^{\binom{m}{2}} t^m}.} \quad (60)$$

The right side has a removable singularity at $t = 0$, with value one. For $|b| > 1$,

$$\mathbb{E}e^{t(X_{1/b} - 1/2)} = \lim_{m \rightarrow \infty} \frac{(-1)^m e^{-t/2} T_{m,b}(\alpha_{m,b} t)}{\alpha_{m,b}^m b^{\binom{m}{2}} t^m}, \quad (61)$$

locally uniformly for complex t .

Proof. The centered moment generating function of (55) is

$$\begin{aligned} \mathbb{E}e^{t(X_b^{[m]} - 1/2)} &= e^{-t/2} \prod_{j=0}^{m-1} \frac{e^{\alpha_{m,b} b^j t} - 1}{\alpha_{m,b} b^j t} \\ &= \frac{(-1)^m e^{-t/2}}{\alpha_{m,b}^m b^{\sum_{j=0}^{m-1} j} t^m} \prod_{j=0}^{m-1} (1 - e^{\alpha_{m,b} b^j t}), \end{aligned}$$

which is (60). The cancellation of the order- m zero is explicit in the product. For $|b| > 1$, [theorem 9.2](#) identifies the finite law with the truncation at $1/b$; bounded support on compact t -sets gives locally uniform convergence of moment generating functions. $\square$

10.2 The four requested reciprocal-radix limits

For $b = 2$, $\alpha_{m,2} = 1/(2^m - 1)$ and $A_2(n) = n$, so

$$\psi_{1/2}(-it) = \mathbb{E}e^{t(X_{1/2} - 1/2)} = \lim_{m \rightarrow \infty} \frac{(-1)^m e^{-t/2}}{\alpha_{m,2}^m 2^{\binom{m}{2}} t^m} \sum_{n=0}^{2^m-1} \varepsilon_n e^{\alpha_{m,2} t n}. \quad (62)$$

Replacing t by it gives a characteristic-function formula. This is a direct limiting representation of the Fabius/Rvachev Fourier product by renormalized Thue–Morse exponential sums.

For $b = -2$, the same theorem converges to $X_{-1/2}$, which is the affine stretch of the Rvachev law in (13). For $b = 4$ and $b = -4$, it converges to the quarter-base laws in (33)–(34). Therefore the eight values named in the prompt are organized into four reciprocal pairs:

exterior radix	reciprocal parameter	Pochhammer base	normalized shape
2	1/2	1/4	Rvachev up
−2	−1/2	1/4	same as up, affinely stretched
4	1/4	1/16	quarter-base plateau profile
−4	−1/4	1/16	same quarter-base profile, affinely stretched

10.3 Why Bernoulli numbers occur on both sides

The normalized factor

$$\frac{e^x - 1}{x}$$

is simultaneously the moment generating function of a uniform digit and the factor obtained after desingularizing $1 - e^x$. Hence the same Bernoulli series (39) governs both geometric-uniform cumulants and the Taylor expansion of Thue–Morse products. More quantitatively,

$$\alpha_{m,b}^{2\ell}[m]_{b^{2\ell}} \longrightarrow \frac{(1 - b^{-1})^{2\ell}}{1 - b^{-2\ell}} \quad (|b| > 1), \quad (63)$$

which is exactly the geometric cumulant factor at $q = 1/b$. Thus the Bernoulli denominators $1 - q^{2\ell}$ are reciprocal-radix limits of finite Gaussian q -integers.

11 A digit-position Thue–Morse/ q -Pochhammer master product

11.1 The two-variable product

Define the binary position weight

$$\pi_2(n) = \sum_{j \geq 0} j b_j(n). \quad (64)$$

For $m \geq 0$, introduce

$$\mathcal{M}_m(u, z; q) = \prod_{j=0}^{m-1} (1 - u q^j z^{2^j}). \quad (65)$$

Theorem 11.1 (Digit-position master identity; classical ingredients, new-to-corpus synthesis). *One has the exact expansion*

$$\mathcal{M}_m(u, z; q) = \sum_{n=0}^{2^m-1} (-u)^{s_2(n)} q^{\pi_2(n)} z^n. \quad (66)$$

Consequently,

$$(u; q)_m = \sum_{n < 2^m} (-u)^{s_2(n)} q^{\pi_2(n)}, \quad (67)$$

and, for each $0 \leq k \leq m$,

$$\sum_{\substack{n < 2^m \\ s_2(n)=k}} q^{\pi_2(n)} = q^{\binom{k}{2}} \begin{bmatrix} m \\ k \end{bmatrix}_q. \quad (68)$$

Proof. Choosing the second term from the j -th factor of (65) records the bit $b_j(n) = 1$. The product of chosen contributions is $(-u)^{s_2(n)} q^{\sum j b_j(n)} z^{\sum 2^j b_j(n)}$, proving (66). Set $z = 1$ for (67). Grouping subsets by cardinality and using the finite q -binomial theorem gives (68); this fixed-weight identity is also a specialization of the q -digital binomial theorem [10]. $\square$

The ordinary Thue–Morse product is obtained by setting $u = 1, q = 1$. The ordinary Hamming-weight enumerator is obtained by setting $q = 1$. The q -Pochhammer symbol is obtained by suppressing the integer coordinate z . Thus (66) simultaneously tracks the integer encoded by a subset, its Hamming weight, and the sum of its occupied bit positions.

11.2 A nondegenerate signed specialization

The direct choice $u = 1, z = 1$ includes the zero factor at $j = 0$. Shift the weight by setting $u = q$ and define

$$\mathcal{T}_m(z; q) = \prod_{j=0}^{m-1} (1 - q^{j+1} z^{2^j}). \quad (69)$$

Let

$$\rho_2(n) = \sum_{j \geq 0} (j+1)b_j(n) = \pi_2(n) + s_2(n).$$

Then

$$\mathcal{T}_m(z; q) = \sum_{n < 2^m} \varepsilon_n q^{\rho_2(n)} z^n, \quad \mathcal{T}_m(1; q) = (q; q)_m. \quad (70)$$

At fixed Hamming weight,

$$\sum_{\substack{n < 2^m \\ s_2(n)=k}} q^{\rho_2(n)} = q^{k(k+1)/2} \begin{bmatrix} m \\ k \end{bmatrix}_q. \quad (71)$$

11.3 Negative q creates a position-parity companion

Let

$$\chi(n) = (-1)^{\pi_2(n)}. \quad (72)$$

For $q = -r$, $r > 0$, the coefficient in (70) becomes

$$\varepsilon_n (-r)^{\rho_2(n)} = (-1)^{s_2(n) + \pi_2(n) + s_2(n)} r^{\rho_2(n)} = \chi(n) r^{\rho_2(n)}. \quad (73)$$

Thus negative q transfers the usual Thue–Morse Hamming parity into parity of the occupied bit positions.

Proposition 11.2 (Four-state automatic companion; DERIVED HERE). *The pair $(\varepsilon_n, \chi(n))$ is 2-automatic. Its recurrences are*

$$\varepsilon_{2n} = \varepsilon_n, \quad \varepsilon_{2n+1} = -\varepsilon_n, \quad (74)$$

$$\chi(2n) = \chi(n)\varepsilon_n, \quad \chi(2n+1) = \chi(n)\varepsilon_n. \quad (75)$$

Proof. Shifting the binary digits of n one place gives $\pi_2(2n) = \pi_2(n) + s_2(n)$; adding the new least significant bit to form $2n+1$ does not change π_2 . The recurrences follow. The pair takes only four sign states, and each input bit updates the state deterministically. $\square$

11.4 Mahler equation and reciprocal symmetry

The infinite product, when convergent as a formal or analytic object, satisfies

$$\mathcal{M}_\infty(u, z; q) = (1 - uz) \mathcal{M}_\infty(uq, z^2; q). \quad (76)$$

The finite product has the exact reciprocity

$$\mathcal{M}_m(u, z; q) = (-u)^m q^{\binom{m}{2}} z^{2^m-1} \mathcal{M}_m(u^{-1}, z^{-1}; q^{-1}). \quad (77)$$

For $\mathcal{T}_m$, this becomes

$$\mathcal{T}_m(z; q) = (-1)^m q^{m(m+1)/2} z^{2^m-1} \mathcal{T}_m(z^{-1}; q^{-1}). \quad (78)$$

These identities give a digit-polynomial counterpart to the reciprocal probability law and reciprocal Lagrange rows.

12 General-radix Bell–Bernoulli formulas

12.1 Exact exponential normal form

Return to $T_{m,b}$ in (59). Put $[m]_b = 1 + b + \dots + b^{m-1}$. The uniform cumulant expansion gives an exact formal normal form.

Theorem 12.1 (General-radix Bell–Bernoulli formula; DERIVED HERE extension). *For every nonzero b ,*

$$T_{m,b}(t) = (-1)^m b^{\binom{m}{2}} t^m \exp \left(\gamma_1 t + \sum_{\ell \geq 1} \gamma_{2\ell} \frac{t^{2\ell}}{(2\ell)!} \right), \quad (79)$$

where

$$\gamma_1 = \frac{1}{2}[m]_b, \quad \gamma_{2\ell} = \frac{B_{2\ell}}{2\ell} [m]_{b^{2\ell}}, \quad \gamma_{2\ell+1} = 0 \quad (\ell \geq 1). \quad (80)$$

Consequently, for every $r \geq 0$,

$$\sum_{S \subseteq \{0, \dots, m-1\}} (-1)^{|S|} \left(\sum_{j \in S} b^j \right)^{m+r} = (-1)^m b^{\binom{m}{2}} \frac{(m+r)!}{r!} \mathcal{B}_r(\gamma_1, \dots, \gamma_r). \quad (81)$$

All corresponding signed power sums of degree below m vanish.

Proof. Factor

$$1 - e^x = -x \frac{e^x - 1}{x}$$

in each term of $T_{m,b}$, and use (39). Summing b^{nj} over j gives $[m]_{b^n}$, yielding (79). Expanding the exponential with complete Bell polynomials and comparing the coefficient of t^{m+r} with the subset-sum expansion (59) gives (81). The factor t^m proves the lower-degree cancellations. $\square$

At $b = 2$, this recovers the dyadic Bell–Bernoulli formulas already developed in the repository. The same theorem supplies explicit versions at $b = -2, 4, -4$, and indeed at arbitrary real or complex radix, without a new calculation.

12.2 Centered positive/negative-radix conjugacy

Theorem 12.2 (Centered radix sign conjugacy; DERIVED HERE). *For $a > 0$,*

$$e^{-\frac{1}{2}[m]_b a t} T_{m,-a}(t) = (-1)^{\binom{m}{2}} e^{-\frac{1}{2}[m]_a t} T_{m,a}(t). \quad (82)$$

Proof. Since $[m]_b = \sum_{j=0}^{m-1} b^j$, the centered product factors as

$$e^{-\frac{1}{2}[m]_b t} T_{m,b}(t) = \prod_{j=0}^{m-1} e^{-b^j t/2} (1 - e^{b^j t}) = \prod_{j=0}^{m-1} (-2 \sinh(b^j t/2)).$$

Replacing $b = a$ by $-a$ multiplies the j -th hyperbolic sine by $(-1)^j$. Their product is $(-1)^{\sum_{j=0}^{m-1} j} = (-1)^{\binom{m}{2}}$. $\square$

This is the discrete signed-transform analogue of theorem 3.1. In the continuous probability law, each centered uniform Fourier factor is a sinc, which is even, so coordinate signs disappear without a residual factor. In the centered Thue–Morse transform, each factor is a hyperbolic sine, which is odd, so the coordinate signs accumulate into the global parity $(-1)^{\binom{m}{2}}$.

13 A unified coordinate-sign and scale-reversal principle

The principal identities can now be arranged into one diagram:

$$\begin{array}{ccc}
 \text{expanding radix } b & \xleftrightarrow{\text{reverse digits}} & \text{contracting ratio } q = b^{-1} \\
 T_{m,b} \text{ with an order-}m \text{ zero} & \xleftrightarrow{\text{desingularize}} & \text{finite uniform convolution } X_b^{[m]} \\
 q\text{-Pochhammer reciprocity} & \longleftrightarrow & \text{geometric Lagrange row reversal} \\
 \text{odd sinh sign accumulation} & \longleftrightarrow & \text{even sinc sign invisibility.}
 \end{array}$$

Three operations repeatedly reappear:

1. **Centering** converts uniform moment factors into even sinc or hyperbolic-sinc factors.
2. **Reversing positions** converts exterior scales $|b| > 1$ into interior ratios $|q| < 1$.
3. **Splitting positions** converts q into q^m , simultaneously dissection of random digits, Gaussian q -integers, and Pochhammer products.

This principle explains why q^2 , rather than q , is the natural base in the centered Fourier Pochhammer factorization: centering makes the factor even, and evenness forgets the sign of every geometric scale.

14 Lambert- W endpoint asymptotics beyond $q = 1/2$

14.1 The positive-parameter Laplace cocycle

For $0 < r < 1$, the Laplace transform is

$$L_r(s) = \mathbb{E}e^{-sX_r} = \prod_{n=0}^{\infty} \frac{1 - e^{-(1-r)r^n s}}{(1-r)r^n s}, \quad s > 0. \quad (83)$$

It satisfies the exact functional equation

$$L_r(s) = \frac{1 - e^{-(1-r)s}}{(1-r)s} L_r(rs). \quad (84)$$

Set

$$\lambda = -\log r, \quad y = \log((1-r)s), \quad G_r(y) = \log L_r\left(\frac{e^y}{1-r}\right).$$

Then

$$G_r(y) - G_r(y - \lambda) = h(y), \quad h(y) = \log \frac{1 - e^{-e^y}}{e^y}. \quad (85)$$

As $y \rightarrow +\infty$, $h(y) = -y + \mathcal{O}(e^{-e^y})$. The quadratic function

$$Q_r(y) = -\frac{y^2}{2\lambda} - \frac{y}{2}$$

satisfies $Q_r(y) - Q_r(y - \lambda) = -y$. This forces the leading log-square behavior behind the known half-base Lambert analysis in the repository.

Conjecture 14.1 (Periodic cocycle normal form; CONJECTURAL). *For every $0 < r < 1$, there is a real-analytic one-periodic function P_r and a constant C_r such that*

$$G_r(y) = -\frac{y^2}{2\lambda} - \frac{y}{2} + C_r + P_r\left(\frac{y}{\lambda}\right) + o(1) \quad (y \rightarrow +\infty), \quad (86)$$

with a full asymptotic expansion obtained by solving the additive cohomological equation (85). The Fourier coefficients of P_r should be expressible through the Mellin transform of $h(y) + y\mathbf{1}_{y>0}$.

The periodic term is not cosmetic. It records the fractional position of the saddle relative to the geometric lattice r^n , exactly as dyadic endpoint asymptotics retain a lower-Lambert phase in the current frontier corpus.

14.2 The Lambert saddle

Ignoring the bounded periodic derivative at first order, the saddle equation for Laplace inversion has the form

$$x \sim \frac{(1-r)y}{\lambda e^y}. \quad (87)$$

Therefore

$$y \sim -W_{-1}\left(-\frac{\lambda}{1-r}x\right), \quad x \downarrow 0, \quad (88)$$

where W_{-1} is the lower real branch of Lambert's function.

Conjecture 14.2 (All-order general- q endpoint transseries; CONJECTURAL). *For each $0 < r < 1$, the density, CDF, and inverse CDF near the left endpoint admit all-order transseries in inverse powers of*

$$w_r(x) = -W_{-1}\left(-\frac{-\log r}{1-r}x\right),$$

whose coefficients are periodic functions of $w_r(x)/(-\log r)$. The coefficient algebra should be generated by Bell polynomials in the derivatives of the cocycle correction P_r , and should reduce at $r = 1/2$ to the existing dyadic Lambert-phase machinery.

14.3 Negative endpoints require no new saddle calculation

Let $a_{-r} = -r/(1-r)$ be the left endpoint of X_{-r} . From (10),

$$F_{-r}(a_{-r} + \delta) = F_r(\delta/c_r). \quad (89)$$

Thus every proved or conjectured positive- r endpoint expansion transfers exactly to the negative parameter. In particular, the relevant Lambert variable is simply $w_r(\delta/c_r)$. The inverse endpoint expansion obeys the affine formula (12). This is substantially stronger than a parallel asymptotic derivation: there is no independent negative- q transseries to compute.

15 Numerical experiments

The accompanying script `q_fabius_experiments.py` is deterministic and contains detailed comments. It uses exact rational arithmetic for finite q -binomial, Lagrange, and digit-product identities; 80-digit arithmetic for analytic product comparisons; and a padded 2^{16} -point FFT for density inversion. The complete output is in `numerical_checks.csv`.

15.1 Selected verification results

identity	parameters	residual	metric
sign conjugacy	$r = 1/2$	0	max char- acteristic error
q -power decimation	$q = 1/2, m = 3$	$6.7 \cdot 10^{-16}$	max char- acteristic error
sinc-Pochhammer logarithm	$q = -1/2, t = 1.1$	$3.7 \cdot 10^{-72}$	80-digit log error
Lagrange reciprocal reversal	all $q = \pm 1/2, \pm 2, \pm 1/4, \pm 4,$ $n = 3, 5, 7$	0	exact rational mismatch count
digit-position product	$m = 8, u = 2/3, q =$ $-1/2$	0	exact coef- ficient mis- matches
standardized sign collapse	$r = 3/4$	$4.8 \cdot 10^{-16}$	max FFT density er- ror
quarter-base plateau	$q = 1/4$	$6.7 \cdot 10^{-16}$	max inter- ior height error

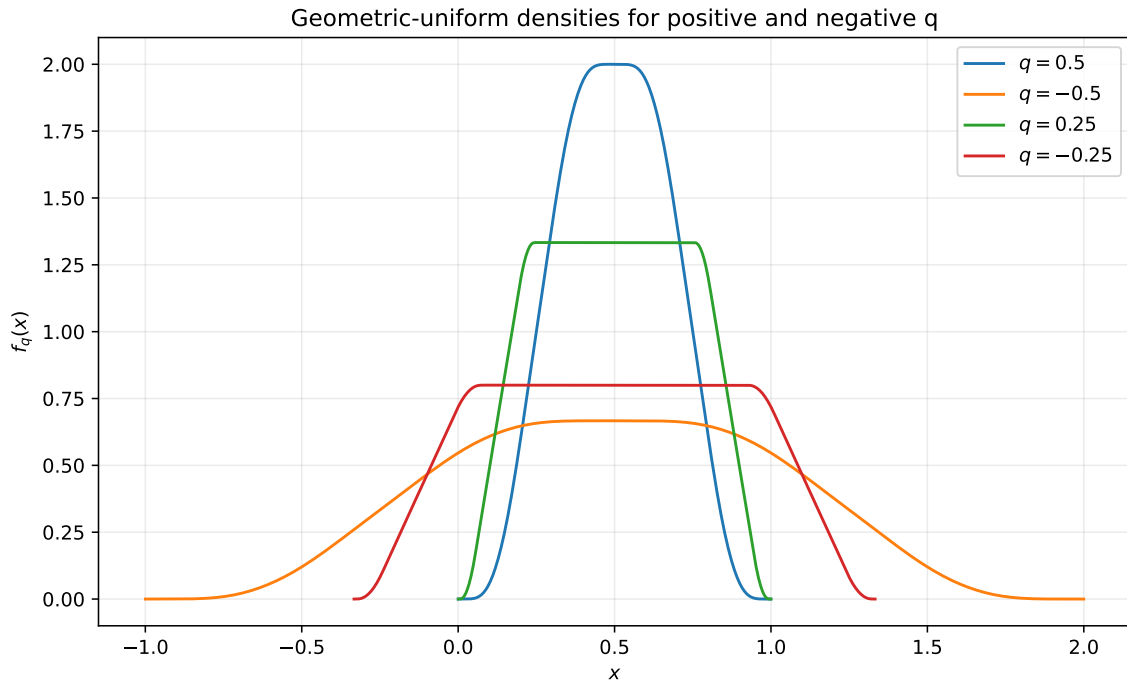


Figure 1: Physical densities for $q = \pm 1/2, \pm 1/4$. The negative curves are exact affine stretches of their positive partners. The quarter-base profiles have positive-length plateaus; the half-base plateau is critical and collapses to the center.

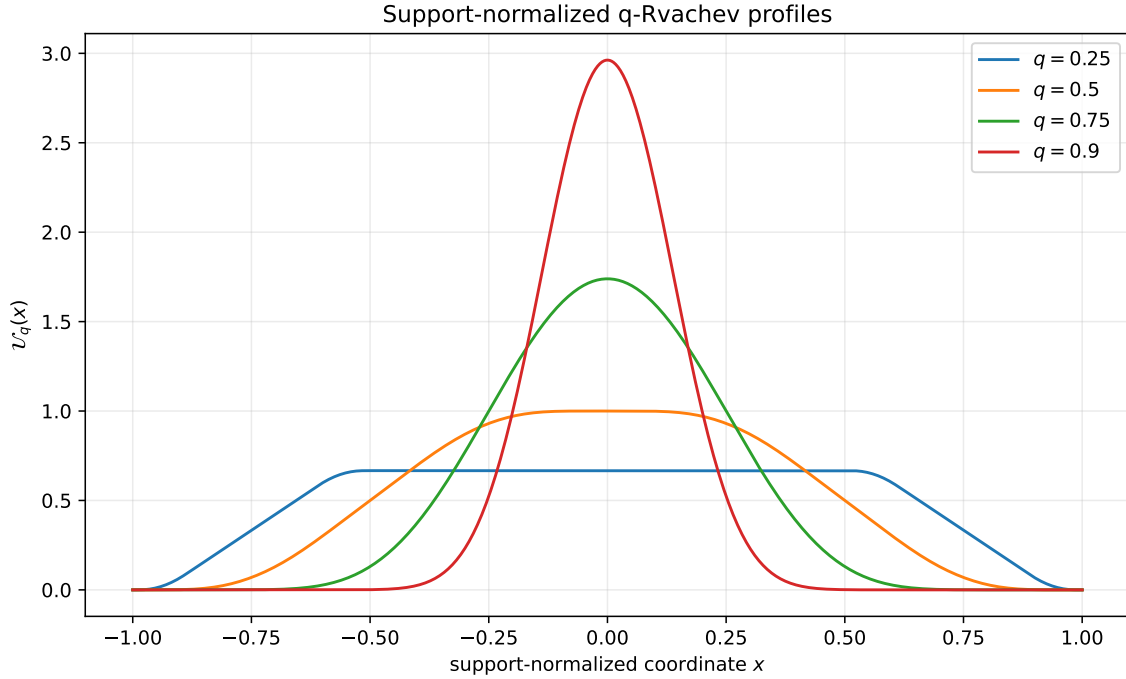


Figure 2: Support-normalized profiles $\mathcal{U}_q$ for positive q . The $q = 1/4$ plateau, the critical Rvachev profile at $q = 1/2$, and the increasingly Gaussian central concentration as $q \rightarrow 1^-$ are visible. The profiles for negative q are identical after support normalization.

15.2 Interpretation and limitations

The numerical work verifies formulas already proved analytically; it is not used as a substitute for proof. FFT inversion is most accurate away from exact support endpoints, where the true density is infinitely flat but floating-point cancellation is severe. The exact plateau and sign-collapse checks are performed on interior grids. The script retains tiny negative values of order machine precision rather than clipping them, so the diagnostic CSV exposes roundoff instead of hiding it.

16 Further conjectures and research directions

16.1 A decimation renormalization semigroup

The maps $q \mapsto q^m$ and weights $a_{m,j}(q)$ satisfy a refinement associativity obtained by splitting indices first modulo m and then modulo n , or directly modulo mn . This suggests a semigroup of exact convolution factorizations acting on densities, cumulants, spectral zeta functions, and Pochhammer products.

Research problem 16.1 (Optimal decimation algorithms). Develop stable exact and numerical algorithms that choose m adaptively to represent a difficult profile at q by simpler profiles at q^m . For $|q|$ near one, repeated powering rapidly enters the plateau regime $|q^m| < 1/2$, where the component density has an exact central constant region. Quantify conditioning, Fourier truncation error, and optimal convolution order.

16.2 Two-base Mahler systems

The master product $\mathcal{M}_m(u, z; q)$ couples a geometric position weight q^j to a lacunary exponent 2^j . Replacing 2 by an integer radix $b \geq 2$ gives

$$\mathcal{M}_\infty^{(b)}(u, z; q) = \prod_{j \geq 0} (1 - uq^j z^{b^j}), \quad \mathcal{M}_\infty^{(b)}(u, z; q) = (1 - uz)\mathcal{M}_\infty^{(b)}(uq, z^b; q).$$

Conjecture 16.2 (Joint natural boundaries; CONJECTURAL). *For generic algebraic (q, z) with $0 < |q| < 1$ and $|z| = 1$, the two-base product has a natural boundary in the z -plane unless the parameters lie on an explicitly classifiable cyclotomic or telescoping locus. The exceptional loci should be controlled by multiplicative dependence between q and powers of z .*

A useful first step is to derive a complete matrix Mahler system for the four-state pair $(\varepsilon_n, \chi(n))$, then study its Dirichlet and Mellin transforms in the spirit of [12].

16.3 Strict shape and total positivity

Log-concavity is only the first layer of possible shape control.

Research problem 16.3 (Pólya-frequency structure). Determine for which q the density f_q , or its sampled Toeplitz kernels, has finite or infinite total positivity. Finite convolutions of interval indicators are box splines, but the geometric limit may retain stronger variation-diminishing properties. Establishing a suitable strict total positivity theorem could resolve strict log-concavity and uniqueness of the mode for $|q| > 1/2$.

16.4 Arithmetic of quarter-base and signed dyadic values

Arias de Reyna and Haugland establish rich arithmetic at dyadic points for the ordinary Fabius function [7, 8]. Decimation suggests that quarter-base values are not merely analogues but convolutional building blocks of the dyadic law.

Research problem 16.4 (Arithmetic transfer through decimation). Investigate whether denominators, valuations, and integrality patterns for $F_{1/2}$ can be decomposed into bilinear or multinomial expressions in values and moments of $F_{1/4}$. The same question at $q = 2^{-m}$ may reveal a hierarchy of arithmetic invariants compatible with Gaussian q -binomial coefficients and the weight vector in (25).

For negative parameters, all value questions reduce affinely to positive parameters. For example, rationality or transcendence of $F_{-1/2}(x)$ is exactly the corresponding question for $F((x+1)/3)$. This should be used to avoid duplicating arithmetic work.

16.5 Complex parameters and roots of unity

The probability interpretation is real, but the finite algebraic identities remain meaningful for complex q . Near roots of unity, denominators in Gaussian interpolation and finite normalizations become singular, while Pochhammer products exhibit cyclotomic cancellation.

Research problem 16.5 (Root-of-unity regularization). Construct canonical renormalized limits of the finite geometric laws, Lagrange rows, and digit-position products as q approaches a root of unity. Classify which singularities are removable after centering, residue-class splitting, or grouping full cyclotomic orbits.

16.6 General- q endpoint analysis

The most promising analytic program is to turn [theorems 14.1](#) and [14.2](#) into theorems. A practical route is:

1. solve [\(85\)](#) by a convergent Mellin–Fourier decomposition after subtracting the quadratic model;
2. obtain uniform bounds for derivatives of the periodic correction;
3. perform the saddle analysis with the periodic phase retained from the start, rather than inserted as a late perturbation;
4. encode the all-order coefficient recursion with complete and partial Bell polynomials;
5. transfer the result to negative q by [\(89\)](#) and to inverse CDFs by affine quantile transport.

17 Formalization roadmap

The exact results in this report are unusually suitable for formalization because their proofs reduce to reindexing summable series, symmetry of the uniform law, finite products, and standard characteristic-function facts. A staged Lean plan is as follows.

17.1 Stage A: sign conjugacy

Formalize the centered uniform symmetry and the deterministic sign-twist of an iid sequence. Prove equality in distribution of the two weighted series, then derive pushforward formulas for CDFs, densities, support, and quantiles. This would close the negative-parameter density gap identified in the current frontier register.

17.2 Stage B: residue-class decimation

Build an equivalence between $\mathbb{N}$ and $\{0, \dots, m-1\} \times \mathbb{N}$, prove absolute summability under reindexing, and establish independence of the subsequences. Characteristic and cumulant corollaries should then be short.

17.3 Stage C: Fourier decay and smoothness

Formalize $|\operatorname{sinc} x| \leq \min(1, 1/|x|)$, the logarithmic choice of the cutoff, and integrability of all polynomial multiples. Reuse the existing Fourier inversion infrastructure to obtain C^∞ densities for negative q and endpoint flatness.

17.4 Stage D: finite algebra

The Pochhammer reciprocity, Gaussian reciprocity, Lagrange reversal, digital master product, and Bell–Bernoulli finite identities are polynomial or finite product statements. They can be proved before any analytic infrastructure. The exact rational checks in the accompanying script supply regression data, not proof certificates.

17.5 Stage E: Thue–Morse desingularization

Define $T_{m,b}$ both as a product and a Boolean-cube sum, prove the order- m zero, and identify the regularized quotient with the finite centered uniform transform. The reciprocal finite-law theorem then supplies the limit at $|b| > 1$.

18 Conclusion

The requested values $\pm 1/2, \pm 1/4, \pm 2, \pm 4$ do not form eight unrelated cases. They sit in a compact web generated by three exact symmetries:

$$q \mapsto -q, \quad q \mapsto q^m, \quad q \mapsto q^{-1}.$$

The first is affine sign conjugacy of centered uniform digits. The second is residue-class decimation and simultaneously Pochhammer dissection. The third is finite digit reversal, Gaussian/Pochhammer reciprocity, and Lagrange row reversal. Their combination yields the following central picture.

- The negative half-base and quarter-base densities are exact affine stretches of the positive profiles; after standardization or support normalization, the sign of q disappears.
- The Fabius/Rvachev law at $q = 1/2$ is an explicit convolution of two quarter-base laws, and more generally a barycenter of $q = 2^{-m}$ laws.
- The centered Fourier transform is a product of q^2 -Pochhammer symbols indexed by the ordinary sine zeros, making the spectral zeta and Bernoulli cumulants immediate.
- Normalized finite laws at $q = 2, -2, 4, -4$ are exactly reciprocal to the interior finite laws and converge to the requested Fabius-type limits.
- Desingularizing the order- m zero of a finite Thue–Morse exponential polynomial produces exactly the centered transform of a finite geometric-uniform convolution. In the reciprocal-radix limit, this becomes the Fabius/Rvachev transform.
- Negative q in the digit-position Pochhammer product replaces the Thue–Morse Hamming-parity sign by a four-state position-parity companion.
- The whole real family is log-concave; $|q| \leq 1/2$ is an exact plateau phase, while $|q| \rightarrow 1$ gives a standardized Gaussian limit.

The strongest next step is a rigorous all- q Lambert- W endpoint theory that retains the geometric periodic phase. The algebraic and probabilistic identities developed here substantially reduce that task: only positive $0 < q < 1$ needs independent asymptotic analysis, and every negative case follows by exact affine transport.

A Detailed derivation of the logarithmic Pochhammer series

For $|t|$ sufficiently small, apply $\log(a; Q)_\infty = -\sum_{m \geq 1} a^m / (m(1 - Q^m))$ to (30):

$$\log \psi_q(t) = - \sum_{k \geq 1} \sum_{m \geq 1} \frac{1}{m(1 - q^{2m})} \left(\frac{(1 - q)^2 t^2}{4\pi^2 k^2} \right)^m \quad (90)$$

$$= - \sum_{m \geq 1} \frac{\zeta(2m)}{m(1 - q^{2m})} \left(\frac{(1 - q)t}{2\pi} \right)^{2m}. \quad (91)$$

Euler's evaluation

$$\zeta(2m) = (-1)^{m+1} \frac{B_{2m}(2\pi)^{2m}}{2(2m)!}$$

turns (91) into

$$\log \psi_q(t) = \sum_{m \geq 1} (-1)^m \frac{B_{2m}}{2m} \frac{(1 - q)^{2m}}{1 - q^{2m}} \frac{t^{2m}}{(2m)!},$$

which is exactly the centered characteristic cumulant series from (40). This calculation is the shortest algebraic route from Pochhammer products to Bernoulli cumulants.

B Selected exact q -Lagrange rows

The archive includes `q_lagrange_weights.csv`, containing exact fractions for $n = 5$ and every $q \in \{\pm 1/2, \pm 2, \pm 1/4, \pm 4\}$. For illustration, at $q = 1/2$,

$$(\lambda_{5,0}, \dots, \lambda_{5,5}) = \left(-\frac{1}{9765}, \frac{2}{315}, -\frac{8}{63}, \frac{64}{63}, -\frac{1024}{315}, \frac{32768}{9765} \right),$$

and the $q = 2$ row is the exact reversal. The alternating and quarter-base rows grow more rapidly, but the reciprocal reversal remains exact over the rationals.

C File inventory and reproducibility

The ZIP supplement contains:

- `fabius_q_connections_report.tex`: this source;
- `fabius_q_connections_report.pdf`: the rendered report;
- `q_fabius_experiments.py`: commented deterministic experiments;
- `numerical_checks.csv`: all residuals and exact mismatch counts;
- `q_lagrange_weights.csv`: exact reciprocal Lagrange rows;
- `standardized_cumulants.csv`: selected closed-form cumulants;
- `physical_densities.pdf/.png` and `normalized_profiles.pdf/.png`: figures;
- `NUMERICAL_README.txt`: execution instructions and package versions.

Run the numerical supplement in its directory with

```
python q_fabius_experiments.py
```

No network access, random seed, or external data file is required.

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